25901163 MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECND200 - FOOD COMMODITIES - I

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

PART A

Objective
Answer all questions
Each question carries 1 mark

1. 50%
2. Soyabean
3. Lycopene
4. Ghee
5. Finger Millet
6. Whole grains
7. Coagulated
8. Unripened
9. Scum
10. Pectic substances

PART B

Short Answer
Answer any 5 questions
Each question carries 2 marks

11. Explain the heat denaturation of whey proteins. (Any 2 points 1 mark each)

- Whey proteins are precipitated or they become insoluble.
- Irreversible reaction
- The randomly coiled proteins open up.
- Amount of coagulum increases with increasing time and temperature of heating.
- The coagulum appears as small particles and collects on the bottom of the pan in which it is heated.

12. Discuss the effect of seed maturity on cooking quality of pulses. ***(Any 2 points 1 mark each)***

- High temperature at the time of maturing affects the cooking time.
- Cooking time increases with the increase in seed maturity.
- Very hard mature seeds take long time to cook.

13. **List out the pigments in fruits.** ***(1/2 mark for each)***

- Chlorophyll, Carotenoids, Anthocyanins, Anthoxanthins

14. Describe the preparation of sweetened condensed milk. (1/2 mark for each point)

- Clarification & pasteurisation of raw milk.
- Sugar is added to increase sweetness and preservation.
- The milk is concentrated by evaporation under vacuum.
- The product is cooled and packed.

15. Define standardised milk and state its fat content. (1 mark for each point)

- It is a mixture of buffalo milk and skim milk
- Fat 4.5% and SNF8.5%

16. Discuss on fortified wheat flour. ***(Any 2 points 1 mark each)***

- Wheat flour is fortified with defatted soya flour (partially gelatinized) up to a maximum of 10%.
- This improves the quantity and quality of protein.
- It also improves functional characteristics such as: Moisture retention and Less oil absorption in the end product
- Iron-fortified wheat flour has been successfully used to prevent iron deficiency anemia.

17. Discuss on glycemic index in pulses. ***(Any 2 points 1 mark each)***

- Beans, peas, lentils, and other pulses have a low glycemic index.
- Legumes are rich in protein, high in fiber, and contain enzyme inhibitors.
- Hence glycemic index is low and therefore and there is lower rise of blood sugar.

Part C

Short Essay
Answer any 6 questions
Each question carries 5 marks

18. **Discuss the mineral and vitamin distribution in rice.** ***(Vitamin 2 marks - each point 1 mark and Minerals 3 marks – each point 1/2 mark)***

- **Vitamins – ***Unpolished and parboiled rice are good sources of B Vitamins. *Rice does not contain either vitamin A or C.
- **Minerals –** *Most of the minerals are located in the pericarp and germ. *Polished rice is poor in calcium and iron. *Coloured types of rice contain more iron than the white rice. *Phosphorous content is high, * 4 percent of which is p[resent as phytic acid. *Rice contains some trace elements.

19. **Explain the occurrence and nutritional significance of trypsin inhibitors in pulses.** ***(Any 5 points 1 mark each)***

- Present in red gram, Bengal gram, cowpea, double beans, soya bean, Lathyrus sativus, and peas.
- Inhibit trypsin – decrease protein digestion.
- Reduce release of essential amino acids (especially methionine).
- Cause pancreatic enlargement and growth retardation (in animals).
- Heat labile.
- Destroyed by moist heat (pressure cooking).
- Autoclaving at 120°C (15–30 min) → complete inactivation.
- Stronger heat needed for soybean and kidney bean.

20. **Discuss the role of green leafy vegetables in preventing deficiency and degenerative diseases. (1 mark for each point)**

- Rich in vitamins (A, C, folic acid) → prevent deficiency diseases like night blindness and anaemia.
- High iron and calcium content → helps prevent anaemia and supports bone health.
- Good source of dietary fibre → aids digestion and prevents constipation.
- Contain antioxidants → reduce risk of degenerative diseases like heart disease and cancer.

- Low in calories and fat → help in preventing obesity and related disorders like diabetes.

21. Explain different types of flavours and off flavours developed in milk.

- **Normal (desirable) flavour (1 mark for each point)**
- Slightly sweet due to lactose
- Flavour sensation in Mouth-Fat, protein and calcium phosphate salts
- **Off Flavours (Any 3 points.1 mark each)**
- Oxidised flavour-oxidation of phospholipids
- off flavour due to auto oxidation of PUFA
- off flavour due to bacterial action, chemical reaction and sunlight exposure.
- Rancid flavour

22. Describe the process of homogenisation and its effect on fat globules in milk. (1 mark for each point)

- Homogenisation- the process of making stable emulsion of milk fat and milk serum.
- Warm milk passed through small aperture under high pressure and velocity.
- Large fat globules are broken into very small, uniform globules.
- The surface area of fat increases and gets evenly distributed.
- Prevents cream separation and improves texture, stability, and digestibility.

23. Describe how water and heat contribute to vitamin loss. (1 mark for each point)

- Water soluble vitamins dissolved in cooking water and get lost.
- Long cooking period increases leaching of water soluble vitamins into water.
- Oxidation vitamin C may be accelerated by heat and causes its loss.
- High temperature destroy heat sensitive vitamins.
- Vitamin A gets oxidised by the application of dry heat.

24. Discuss on evaporated milk and dry milk.

Evaporated milk (1 mark for each point)

- 60-70% water from milk has been evaporated.
- Raw milk-clarified-concentrated-fortified-homogenised-sterilised –cooled

Dry Milk (1 mark for each point)

- Milk - dehydrated to about 97% by spray drying or vacuum drying.

- It can be reconstituted into fluid milk.
- It has good shelf life without refrigeration.

25. Discuss on barley. ***(Any 5 points 1 mark each)***

- Belongs to genus Hordeum vulgare.
- Barley protein has low lysine content; hence its biological value is lower than wheat.
- It contains little gluten, so it is not suitable for making good quality bread or chapatis.
- Barley is processed into: Shelled barley, Pearl barley, Barley grits, Barley flakes, Barley flour
- The grain has a fused husk, and pearling involves: Dehusking, Removal (abrasion) of bran layers
- Pearl barley is: Used in soups and infant foods, also used for medicinal purposes
- Barley is cultivated under both: Irrigated and non-irrigated conditions
- Major use of barley: *Brewing and distilling industry *Used for production of beer and whiskey
- Barley malt is used in bakery, processed foods and in vinegar and syrup making.
- Sprouted barley – *amylase activity is increased. *Also known as Malt used in brewing industry to help digest starchy material into sugar for rapid yeast fermentation.
- Malt also contributes to the flavour of brewed beverages such as beer.

Part D

Essay
Answer any 2 questions
Each question carries 10 marks

26. Explain the composition and nutritive value of cereals.

- **Energy-***main source of energy *contributing 70-80% of requirement. *340 Kcal of energy per 100 g. ***(1 mark)***
- **CHO-** 8*0% of dry matter is CHO, *2 CHO present – crude fibre and soluble CHO. *Small quantities of dextrin and sugars are also present. *Free sugars present are simple sugars such as glucose, sucrose, maltose. ***(1 marks)***

- **Protein-** *Rice contains less amount of protein compare to other cereals. *Protein found in tissues of cereal grain. *High concentration in embryo, scutellum and aleurone layer than in endosperm. * Within the endosperm the concentration of protein increases from the centre to the periphery. *Types of protein-albumins, globulins, prolamines and glutelins. *Contain 7-11% of protein. *Generally deficient in lysine. *Provide 50% of protein requirement. ***(2 marks)***
- **Fats-** 1-2% in wheat and rice, 3% in maize. More fats are present in germ and bran. *Wheat germ contains lipids 6-11% and bran 3-5% and endosperm 0.8-1.5%. *Fats are mostly the triglycerides of palmitic, oleic and linoleic acid.
  (2 marks)
- **Minerals-** About 95 percent of minerals are the phosphates and sulphates of potassium, magnesium and calcium. Considerable part of phosphorous in cereals is present in the form of phytin. *Some mineral elements like copper,zinc and manganese are also present in very small quantities. * poor source of calcium and iron particularly rice. *Ragi rich source of Calcium and iron. *Millets are rich in minerals and fibre. ***(2 marks)***
- **Vitamins-** Whole grain are important source of B vitamins. *Since most of the vitamins are in outerbran - refining or polishing the grains reduce B vitamins. *Milled and parboiled rice retain much of B vitamins. *Maida has less B vitamin. ***(1 marks***
- **Enzymes-** *Contain many enzymes- amylases, proteases, lipases and oxido reductase. *Upon germination alpha-amylase activity increases. *Proteases more in germ. ***(1 mark)***

27. Discuss the processing of pulses, including soaking, germination, fermentation, parching and extrusion.

Soaking- ***(2 marks)***

- Pulses with hard outer covering need soaking before cooking.
- Water enters through the hilum and causes swelling of cotyledons.
- Makes pulses tender and reduces cooking time.
- Moisture content increases from 10% to 60–70%.
- Reduces phytic acid and oligosaccharides.

- Salt solution soaking improves water absorption and loosens seed coat.
- Soda reduces cooking time, but causes loss of thiamine.
- Hard water delays cooking.
- Causes leaching of water-soluble nutrients.
- Soaking water can be used in cooking to reduce nutrient loss.

Germination- ***(2 marks)***

- Pulses are soaked overnight, drained, and kept moist for sprouting.
- Requires moisture and warmth; occurs in 1–2 days.
- Green gram germinates faster; process is quicker in summer.
- Enzymes are activated during germination.
- Starch → sugars, proteins → simpler forms (improves digestibility).
- Reduces trypsin inhibitors and phytic acid.
- Increases vitamins (especially B-complex) and Vitamin C is synthesized.
- Improves availability of minerals like iron, calcium, and zinc.
- Reduces cooking time and makes seeds softer.
- Decreases flatulence.
- Improves taste, texture, and palatability.

Fermentation- ***(2 marks)***

- Fermentation involves microbial action that breaks down pulse components.
- Increases digestibility by breakdown of proteins.
- Improves palatability and cooking quality.
- Enhances nutritive value of pulses.
- Improves availability of essential amino acids.
- Helps in elimination of toxic substances (e.g., in idli preparation).
- Improves overall protein quality.
- Common examples: idli and dosa.

Parching- ***(2 marks)***

- Parching is a dry heat treatment used to improve acceptability of pulses.
- Common pulses: Bengal gram and peas.
- Grains are first sprinkled with a little water.
- Then mixed with preheated sand in a frying pan.
- Heated on open fire at 200–250°C for 2–3 minutes.
- Produces roasted, ready-to-eat products with good flavor.
- Improves palatability and digestibility.

Extrusion- ***(2 marks)***

- Blends of cereals and legume flour are extruded at high temperature (140–200°C) high pressure and moisture content <20%.
- The product is cooked for a short time, (30 to 60 seconds), and the product is forced out of the extruder through a variety of dies and dried on conveyor belts.

28. Explain classification of fruits and its nutritive value.

Classification – ***(5 marks)***

- Berries – Strawberries, Gooseberries, Raspberries
- Citrus fruits – Oranges, Lime, Lemon
- Drupes- Apricot, Peach, Plum
- Grapes- Green grapes
- Melons- Musk melon, water melon
- Tropical and subtropical fruits – Avocado, Banana, Dates, Guava

Nutritive value - ***(5 marks)***

- *Poor source of protein and fat. *Avocado is the exception containing 13.8 percent fat. *Omega-3-fatty acid present in plum, papaya and guava.
- High amount moisture – highly perishable.
- Good source of fibre.
- Not good source of calories. *Bananas give fairly good amount of calories.
- Ripe fruits contain high sugar than unripe fruit (Sugar in the form of sucrose, fructose and glucose).
- Poor source of iron.
- Mangoes excellent source of carotenes.
- Guava and citrus fruits are good source of vitamin C.
- Most fruit in raw state contain some ascorbic acid.
- Gooseberry is the richest source of vitamin C.
- Apple, pears, cherries, grapes and citrus fruits contain flavonoids which act as antioxidants.

29. Discuss in detail the nutritive value of milk (1 mark for each point)

- Milk has high quality protein with a biological value over 90. Lysine is the essential amino acid present in milk.
- Milk is the only substance that contains lactose which has galactose essential for the synthesis of myelin sheath.
- Lactose favours the absorption of calcium.
- Lactose has controlled glycaemic index, so it is preferred as a source off carbohydrate.
- Milk fat is easily digestible and it contains essential fatty acids.
- Milk is an excellent source of calcium. The calcium requirement cannot be met easily without taking milk.
- The calcium phosphorus ratio in milk is more favourable for bone development.
- The iron present in milk is readily absorbed by our body.
- Milk is good source of Vitamin A & D.
- It is a good source of riboflavin and niacin and a fair source of thiamine.

